LATTICES OF SUBVARIETIES AND OF LOGICS AS SUBLATTICES OF THE LATTICE OF TWO-OUT-OF-THREE CLASSES

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Abstract. For a category C let $\mathrm{Same}(C)$ be the complete lattice of classes of morphisms that contain the identities and are closed under composition and two-out-of-three. Every reflective subcategory $\mathcal{R}$ contributes the element $W_{\mathcal{R}}$ of maps inverted by the reflector. We study this assignment on the lattice $\mathrm{Sub}(\mathcal{V})$ of subvarieties of a variety $\mathcal{V}$ of algebras. The meet of two subvarieties always goes to the join of the two classes, $W_{V\wedge V'} = W_V \vee W_{V'}$, because the reflector into $V \wedge V'$ is the composite of the two reflectors; for Grothendieck topologies the corresponding statement requires towers of plus constructions and can fail. The join of two subvarieties goes to the meet of the classes, $W_{V\vee V'} = W_V \cap W_{V'}$, exactly when the two reflectors jointly reflect isomorphisms on $V \vee V'$, and we characterize this: every algebra of $V \vee V'$ must be a subdirect product of its two reflections, and no algebra of $V \vee V'$ may have a proper subalgebra dense for both reflectors. The first condition holds in every congruence distributive variety by Jónsson's lemma, the second in every congruence permutable variety by a Chinese remainder argument; so for arithmetical varieties $V \mapsto W_V$ is a lattice embedding of $\mathrm{Sub}(\mathcal{V})^{\mathrm{op}}$ into $\mathrm{Same}(\mathcal{V})$. Neither hypothesis can be dropped: the modular group of order 16 gives a counterexample in groups, and the three-element chain inside the four-element Boolean lattice gives one in bounded lattices with a constant. Through algebraizability the lattices of normal modal logics, of intermediate logics, of substructural logics over **FL** and of fuzzy logics are therefore sublattices of the sameness lattices of their algebras. A lattice embedding of $\mathrm{Sub}(\mathcal{V})^{\mathrm{op}}$ forces every subdirectly irreducible algebra to generate a join-prime subvariety and hence $\mathrm{Sub}(\mathcal{V})$ to be distributive; so the lattice of λ-theories, which is not modular, embeds as a join-semilattice but not as a lattice: over the pentagon of Berarducci-tree theories of Severi and de Vries, the quotient map between two term algebras is inverted by two reflections but not by the reflection into their meet.

2020 *Mathematics Subject Classification.* 08B15 (primary), 08B10, 18A40, 03B45, 03B47, 03B40 (secondary).

Key words and phrases. variety, subvariety lattice, verbal congruence, reflective subcategory, two-out-of-three, congruence distributive, congruence permutable, Jónsson's lemma, lattice of modal logics, substructural logic, lambda theories.

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1. Introduction

A *sameness* on a category C is a class of morphisms containing the identities and closed under composition and two-out-of-three; the samenesses form a complete lattice $\mathrm{Same}(C)$ under inclusion, with meets given by intersection and joins by closure [13]. Every functor F out of C gives a sameness $W_F = \{f : Ff \text{ invertible}\}$, and in earlier work the elements W_{a_J} contributed by the sheafification functors a_J of the Grothendieck topologies J on a small category were studied: $J \mapsto W_{a_J}$ preserves all meets, preserves binary joins for every finite category, and fails to preserve them on the poset of natural numbers [16, 17]. The positive results are proved by showing that alternating towers of plus constructions, or of separated quotients, connect a presheaf to its sheafification for the join topology inside the join of the two classes.

This paper carries out the same programme for the other classical lattice of reflective subcategories: the lattice $\mathrm{Sub}(\mathcal{V})$ of subvarieties of a variety $\mathcal{V}$ of algebras. Each subvariety V is reflective, with reflector $a_V(A) = A/\theta_V(A)$ for the verbal congruence $\theta_V(A)$, and we write $W_V := W_{a_V} = \{f : a_V f \text{ is an isomorphism}\}$. The two lattice operations behave differently.

Theorem 1.1. *For every variety $\mathcal{V}$ and all subvarieties V, V', $W_{V\wedge V'} = W_V \vee W_{V'}$ in $\mathrm{Same}(\mathcal{V})$.*

Theorem 1.2. *For subvarieties V, V' of any variety, $W_{V\vee V'} = W_V \cap W_{V'}$ if and only if*

(i) *every algebra of $V \vee V'$ is a subdirect product of its V-reflection and its V'-reflection, and*
(ii) *no algebra of $V \vee V'$ has a proper subalgebra that is both V-dense and V'-dense, where a subalgebra is V-dense if its inclusion is inverted by the V-reflector.*

Condition (ii) holds whenever the verbal congruences $\theta_V(A)$ and $\theta_{V'}(A)$ permute for every $A \in V \vee V'$; together with (i) this says that A is the pullback of its V- and V'-reflections over its $V \wedge V'$-reflection.

Corollary 1.3. *If $\mathcal{V}$ is congruence distributive and congruence permutable, then $W_{V\vee V'} = W_V \cap W_{V'}$ for all subvarieties V, V', and $V \mapsto W_V$ is a lattice embedding $\mathrm{Sub}(\mathcal{V})^{\mathrm{op}} \hookrightarrow \mathrm{Same}(\mathcal{V})$. Neither hypothesis can be omitted: the equality fails for the subvarieties $\mathrm{var}(\mathbb{Z}_8)$ and $\mathrm{var}(D_8)$ of groups (Example 4.3), and for the subvarieties $c = 0$ and $c = 1$ of bounded lattices with a constant c (Example 4.4); permutability, on the other hand, is not necessary (Example 4.5).*

Theorem 1.4. *For subvarieties V, V' of a variety $\mathcal{V}$, condition (i) of Theorem 1.2 holds if and only if every subdirectly irreducible algebra of $V \vee V'$ lies in $V \cup V'$. Consequently, if $W_{V\vee V'} = W_V \cap W_{V'}$ for all pairs of subvarieties, then every subdirectly irreducible algebra of $\mathcal{V}$ generates a join-prime element of $\mathrm{Sub}(\mathcal{V})$, and $\mathrm{Sub}(\mathcal{V})$ is distributive. More precisely, if $\mathrm{Sub}(\mathcal{V})$*

contains a pentagon $P \subsetneq Q$, R with $P \vee R = Q \vee R$ and $P \wedge R = Q \wedge R$, then $W_{P \vee R} \neq W_P \cap W_R$.

Theorem 1.1 is elementary: because a subvariety is closed under quotients, the reflector into $V \wedge V'$ is the composite $a_{V'}a_V$, so the unit of $V \wedge V'$ is a composite of a V-unit and a V'-unit and the join theorem of [15] applies. The contrast with sheaf theory is the point: sheafification for J' does not preserve J-sheaves, the reflector into $J \vee J'$ is not a composite of the two, and the towers of [16, 17] are the substitute, which exists for finite categories and does not exist on $\mathbb{N}$. Theorem 1.2 is where the algebra enters. A map g of $V \vee V'$ lies in $W_{V \vee V'}$ only if it is an isomorphism, while $g \in W_V \cap W_{V'}$ only sees the two reflections of g; so the question is whether the pair of reflectors $(a_V, a_{V'})$ reflects isomorphisms on $V \vee V'$, i.e. whether the "fracture" functor $V \vee V' \to V \times_{V \wedge V'} V'$ of Figure 2 is conservative. Condition (i) is injectivity, (ii) surjectivity. In a congruence distributive variety (i) holds for all pairs by Jónsson's lemma [6], since the subdirectly irreducible algebras of $V \vee V'$ lie in $V \cup V'$; in a congruence permutable variety (ii) holds by a Chinese remainder argument. In an arithmetical variety every algebra of $V \vee V'$ is therefore the pullback of its V- and V'-reflections over its $V \wedge V'$-reflection, the algebraic analogue of a fracture square.

What is known. The reflector of a subvariety, verbal congruences, Jónsson's lemma and the Mal'cev condition for permutability are standard [3]. That a join of two subvarieties of a congruence permutable variety is the direct product of the two when they are disjoint is due to Jónsson and Tsinakis; joins that are subdirect products are studied by Kowalski and Paoli [8]. The lattice of samenesses, the join theorem and the results for Grothendieck topologies are from [13, 15, 16, 17].

What is new. The main result is Theorem 1.2: an exact characterization of the pairs of subvarieties whose reflectors jointly reflect isomorphisms, with condition (i) the injectivity and condition (ii) the surjectivity of the comparison map of Figure 2; the arithmetical case (Corollary 1.3) is the instance where both conditions hold for all pairs, and the examples show that each condition can fail on its own. Further new are Theorem 1.1, Theorem 1.4, the three examples, Proposition 4.7, which shows that the equality is not a property of the lattice $\mathrm{Sub}(\mathcal{V})$ alone, and the consequences for logic (Section 5): since the lattice of axiomatic extensions of an algebraizable logic is dually isomorphic to the subvariety lattice of its equivalent algebraic semantics [1], the lattices of normal modal logics, of intermediate logics, of substructural logics over **FL** and of the usual fuzzy logics are sublattices of the sameness lattices of modal algebras, Heyting algebras, **FL**-algebras and their subvarieties (Corollary 5.1); the lattice of λ-theories, which is isomorphic to the lattice of equational theories of lambda abstraction algebras [10], embeds as a join-semilattice (Corollary 5.3) but, by Theorem 1.4 and

the non-modularity of the lattice of λ-theories [11], not as a lattice (Corollary 5.4); over the explicit pentagon of λ-theories found by Severi and de Vries [12], the failure is the quotient map of term algebras of Example 5.7. Theorem 1.4 is the observation that condition (i) is a statement about subdirectly irreducible algebras, so that the lattice embedding, which holds for arithmetical varieties by Jónsson's lemma, conversely imposes Jónsson's conclusion on the subdirectly irreducibles of every join and thereby distributivity on $\mathrm{Sub}(\mathcal{V})$.

2. Preliminaries

2.1. **Samenesses.** For a category C, $\mathrm{Same}(C)$ is the set of samenesses ordered by inclusion; it is a complete lattice, $U \wedge U' = U \cap U'$, and $U \vee U'$ is the smallest sameness containing $U \cup U'$ [13]. For a functor $P\colon C \to C$ put $W_P = \{f : Pf \text{ invertible}\}$; it is a sameness containing all isomorphisms.

Theorem 2.1 (join theorem, [15, Thm. 4.1])**.** *Let U, V be samenesses on C whose join contains the isomorphisms, and let $P\colon C \to C$ be a functor with a natural transformation $\pi\colon \mathrm{id} \Rightarrow P$ such that $\pi_X \in U \vee V$ for all X and P inverts U and V. Then $U \vee V = W_P$.*

Proof. $W_P \supseteq U \vee V$ since W_P is a sameness containing U and V. If Pf is invertible then $Pf \in U \vee V$, so the naturality square $\pi_Y f = Pf\, \pi_X$ has three sides in $U \vee V$, hence the fourth. □

2.2. **Subvarieties and verbal congruences.** Let $\mathcal{V}$ be a variety of algebras of a fixed signature and $\mathrm{Sub}(\mathcal{V})$ its lattice of subvarieties; $V \wedge V' = V \cap V'$ and $V \vee V'$ is the variety generated by $V \cup V'$. For $A \in \mathcal{V}$ and $V \in \mathrm{Sub}(\mathcal{V})$ the *verbal congruence* $\theta_V(A)$ is the smallest congruence of A whose quotient lies in V; it is the congruence generated by the pairs $(s(\bar{a}), t(\bar{a}))$ for the identities $s = t$ of V and tuples $\bar{a}$ of A, and a homomorphism $f\colon A \to B$ satisfies $f(\theta_V(A)) \subseteq \theta_V(B)$. Thus $a_V(A) = A/\theta_V(A)$ is a reflector of $\mathcal{V}$ onto V with unit the quotient map $u_A\colon A \to a_V A$, and we put

$$W_V := \{f \in \mathrm{Mor}(\mathcal{V}) : a_V f \text{ is an isomorphism}\} \in \mathrm{Same}(\mathcal{V}).$$

If $V \subseteq V'$ then $a_V = a_V a_{V'}$ and $W_{V'} \subseteq W_V$. The map $V \mapsto W_V$ is an order-embedding $\mathrm{Sub}(\mathcal{V})^{\mathrm{op}} \hookrightarrow \mathrm{Same}(\mathcal{V})$: if $V \not\subseteq V'$, pick $A \in V \setminus V'$; its V'-unit is in $W_{V'}$, not an isomorphism, and a_V sends it to itself (as $a_V A = A$ and $a_V(A/\theta_{V'}(A)) = A/\theta_{V'}(A)$ because both lie in V, which is closed under quotients), so it is not in W_V. We write $\mathrm{Mor}(\mathcal{V})$ for the class of homomorphisms between algebras of $\mathcal{V}$.

Lemma 2.2. *For $V, V' \in \mathrm{Sub}(\mathcal{V})$ and $A \in \mathcal{V}$ the composite $A \to A/\theta_V(A) \to (A/\theta_V(A))/\theta_{V'}(A/\theta_V(A))$ is the $V \wedge V'$-unit of A; that is, $a_{V \wedge V'} = a_{V'} a_V$ and $\theta_{V \wedge V'}(A) = \theta_V(A) \vee \theta_{V'}(A)$.*

Proof. The target lies in V' by construction and in V because V is closed under quotients. Any homomorphism $A \to C$ with $C \in V \cap V'$ factors

through $A/\theta_V(A)$ since $C \in V$, and the factor through $(A/\theta_V(A))/\theta_{V'}$ since $C \in V'$. By the correspondence theorem the kernel of the composite is $\theta_V(A) \vee \theta_{V'}(A)$. □

3. Meets of subvarieties

Proof of Theorem 1.1. Apply Theorem 2.1 with $U = W_V$, $V = W_{V'}$ (the join contains the isomorphisms because each does), $P = a_{V\wedge V'}$ and π the unit. By Lemma 2.2, π_A is the composite of the V-unit of A, which lies in W_V because a_V of a unit is an isomorphism, and the V'-unit of $A/\theta_V(A)$, which lies in $W_{V'}$; so $\pi_A \in W_V \vee W_{V'}$. Since $a_{V\wedge V'} = a_{V\wedge V'}a_V = a_{V\wedge V'}a_{V'}$, the functor $a_{V\wedge V'}$ inverts W_V and $W_{V'}$. Hence $W_V \vee W_{V'} = W_{a_{V\wedge V'}} = W_{V\wedge V'}$. □

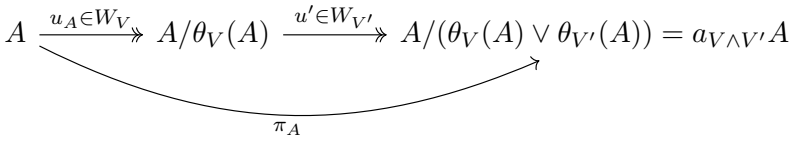


Figure 1. The unit of $V \wedge V'$ is a composite of two units (Lemma 2.2); this single step replaces the towers needed for Grothendieck topologies.

The argument uses only that each subvariety is closed under the reflections into the other; it holds verbatim for reflective subcategories $\mathcal{R}, \mathcal{R}'$ of any category with $a_{\mathcal{R}'}(\mathcal{R}) \subseteq \mathcal{R}$, where then $a_{\mathcal{R}\cap\mathcal{R}'} = a_{\mathcal{R}'}a_{\mathcal{R}}$ and $W_{\mathcal{R}} \vee W_{\mathcal{R}'} = W_{\mathcal{R}\cap\mathcal{R}'}$. Full subcategories of sheaves do not have this property: for two Grothendieck topologies J, J' the J'-sheafification of a J-sheaf need not be a J-sheaf, $a_{J\vee J'}$ is not the composite $a_{J'}a_J$, and the join $W_{a_J} \vee W_{a_{J'}}$ is the class $W_{a_{J\vee J'}}$ only when a tower of alternating constructions reaches the join sheafification, which it does for finite categories [17] and does not on the poset $\mathbb{N}$ [16, Thm. 7.1].

4. Joins of subvarieties

The inclusion $W_{V\vee V'} \subseteq W_V \cap W_{V'}$ always holds, since $a_V = a_V a_{V\vee V'}$ and $a_{V'} = a_{V'}a_{V\vee V'}$. For the converse we reduce to maps inside $V \vee V'$.

Lemma 4.1. *$W_{V\vee V'} = W_V \cap W_{V'}$ holds if and only if every homomorphism $g\colon A \to B$ with $A, B \in V \vee V'$ for which $a_V g$ and $a_{V'} g$ are isomorphisms is itself an isomorphism.*

Proof. If $f \in W_V \cap W_{V'}$, then $g := a_{V\vee V'} f$ is a map in $V \vee V'$ with $a_V g = a_V f$ and $a_{V'} g = a_{V'} f$ isomorphisms; if the condition holds, g is an isomorphism and $f \in W_{V\vee V'}$. Conversely, a map g in $V \vee V'$ has $a_{V\vee V'} g = g$, so $g \in W_{V\vee V'}$ iff g is an isomorphism. □

Definition 4.2. A subalgebra $C \subseteq B$ is *V-dense* if the inclusion lies in W_V, i.e. if C meets every class of $\theta_V(B)$ and $\theta_V(B) \cap C^2 = \theta_V(C)$.

Proof of Theorem 1.2. Suppose (i) and (ii) and let $g\colon A \to B$ be a map in $V \vee V'$ with $a_V g$, $a_{V'} g$ isomorphisms; by Lemma 4.1 we must show that g is bijective. If $g(x) = g(y)$ then $x\,\theta_V(A)\,y$ and $x\,\theta_{V'}(A)\,y$ because $a_V g$ and $a_{V'} g$ are injective, so $x = y$ by (i): g is injective. Its image $C := g(A)$ is a subalgebra of B with $C \cong A$, and the inclusion $C \hookrightarrow B$ is inverted by a_V and $a_{V'}$ (it is g composed with an isomorphism); so C is V-dense and V'-dense and $C = B$ by (ii).

Conversely suppose $W_{V \vee V'} = W_V \cap W_{V'}$. If (i) fails for some $A \in V \vee V'$, let $\theta := \theta_V(A) \cap \theta_{V'}(A) \neq \Delta$ and $g\colon A \to B := A/\theta$ the quotient map; $B \in V \vee V'$ as a subdirect product of $A/\theta_V(A) \in V$ and $A/\theta_{V'}(A) \in V'$. By the correspondence theorem $\theta_V(B) = (\theta_V(A) \vee \theta)/\theta = \theta_V(A)/\theta$, so $a_V g$ is the identity of $A/\theta_V(A)$, and likewise for V'; thus $g \in W_V \cap W_{V'}$, but g is not injective, contradicting Lemma 4.1. If (ii) fails, a V-dense and V'-dense proper subalgebra $C \subsetneq B$ with $B \in V \vee V'$ gives an inclusion in $W_V \cap W_{V'}$ which is not an isomorphism, again contradicting Lemma 4.1.

Finally let the verbal congruences permute on every algebra of $V \vee V'$ and let $C \subseteq B$ be V-dense and V'-dense, $B \in V \vee V'$; we show $C = B$. Write $\theta_V, \theta_{V'}$ for the verbal congruences of B. Given $b \in B$, density gives $c_1, c_2 \in C$ with $c_1\,\theta_V\,b$ and $c_2\,\theta_{V'}\,b$. Then c_1 and c_2 are related by $\theta_V \vee \theta_{V'} = \theta_{V \wedge V'}$ (Lemma 2.2), hence, as $a_{V \wedge V'} = a_{V'} a_V$ inverts the inclusion, by $\theta_{V \wedge V'}(C) = \theta_V(C) \vee \theta_{V'}(C)$ inside C; by permutability in C (also an algebra of $V \vee V'$) there is $c \in C$ with $c_1\,\theta_V(C)\,c\,\theta_{V'}(C)\,c_2$. Then $c\,\theta_V\,c_1\,\theta_V\,b$ and $c\,\theta_{V'}\,c_2\,\theta_{V'}\,b$, so $c\,(\theta_V \cap \theta_{V'})\,b$ and $c = b$ by (i) for B. □

$$\begin{array}{ccc} A & \twoheadrightarrow & A/\theta_V(A) \\ \downarrow & \lrcorner & \downarrow \\ A/\theta_{V'}(A) & \twoheadrightarrow & A/(\theta_V(A) \vee \theta_{V'}(A)) \end{array}$$

FIGURE 2. The fracture square of $A \in V \vee V'$. Condition (i) of Theorem 1.2 is injectivity of the comparison map $A \to A/\theta_V \times_{A/(\theta_V \vee \theta_{V'})} A/\theta_{V'}$, and permutability of $\theta_V(A)$, $\theta_{V'}(A)$ is its surjectivity; the square is a pullback for every A in an arithmetical variety.

Proof of Theorem 1.4. Suppose (i) and let $A \in V \vee V'$ be subdirectly irreducible. Then $\theta_V(A) \cap \theta_{V'}(A) = \Delta$, and Δ is meet-irreducible in the congruence lattice of A, so $\theta_V(A) = \Delta$ or $\theta_{V'}(A) = \Delta$, i.e. $A \in V$ or $A \in V'$. Conversely suppose every subdirectly irreducible algebra of $V \vee V'$ lies in $V \cup V'$ and let $A \in V \vee V'$. Writing A as a subdirect product of subdirectly irreducible quotients A/κ_i (Birkhoff), each κ_i contains $\theta_V(A)$ or $\theta_{V'}(A)$,

hence contains $\theta_V(A) \cap \theta_{V'}(A)$; as $\bigcap_i \kappa_i = \Delta$, so is $\theta_V(A) \cap \theta_{V'}(A)$, which is (i).

Now let $W_{V \vee V'} = W_V \cap W_{V'}$ for all pairs, so that (i) holds for all pairs by Theorem 1.2. If A is subdirectly irreducible and $\mathrm{var}(A) \subseteq V \vee V'$, then $A \in V \vee V'$, so $A \in V$ or $A \in V'$, i.e. $\mathrm{var}(A) \subseteq V$ or $\mathrm{var}(A) \subseteq V'$: $\mathrm{var}(A)$ is join-prime. For distributivity let $X, Y, Z \in \mathrm{Sub}(\mathcal{V})$; the variety $(X \vee Y) \wedge Z$ is generated by its subdirectly irreducible members, each of which lies in X or in Y and in Z, hence in $(X \wedge Z) \vee (Y \wedge Z)$; so $(X \vee Y) \wedge Z \subseteq (X \wedge Z) \vee (Y \wedge Z)$, and the reverse inclusion always holds.

Finally let $P \subsetneq Q$, R be a pentagon. Since $Q \not\subseteq P$, some subdirectly irreducible $A \in Q$ is not in P; then $A \in Q \subseteq Q \vee R = P \vee R$, and $A \notin R$, for otherwise $A \in Q \wedge R = P \wedge R \subseteq P$. So (i) fails for the pair (P, R) by the first part, and $W_{P \vee R} \neq W_P \cap W_R$ by Theorem 1.2. □

Proof of Corollary 1.3. Let $\mathcal{V}$ be congruence distributive and congruence permutable. The verbal congruences permute, so (ii) holds by Theorem 1.2. For (i), by Jónsson's lemma [6] every subdirectly irreducible algebra of $V \vee V'$ lies in $HSP_u(V \cup V')$; an ultraproduct of algebras each in V or in V' lies in V or in V' (one of the two index sets belongs to the ultrafilter), and V, V' are closed under H and S, so the subdirectly irreducibles of $V \vee V'$ lie in $V \cup V'$, and (i) holds by Theorem 1.4. Theorem 1.2 gives $W_{V \vee V'} = W_V \cap W_{V'}$, and with Theorem 1.1 and the order-embedding of Section 2 the map $V \mapsto W_V$ is a lattice embedding. The examples below show that the hypotheses cannot be dropped and that permutability is not necessary. □

Example 4.3 (permutable, not distributive)**.** In the variety of groups let $V = \mathrm{var}(\mathbb{Z}_8)$, the abelian groups of exponent dividing 8, and $V' = \mathrm{var}(D_8)$, the variety generated by the dihedral group of order 8. Let $M_{16} = \langle a, b \mid a^8 = b^2 = 1,\ bab = a^5 \rangle$ be the modular group of order 16. It lies in $V \vee V'$: in $\mathbb{Z}_8 \times D_8$, with r a rotation of order 4 and s a reflection, the subgroup K generated by $x = (1, r)$ and $y = (0, s)$ contains $x \cdot yxy = (1, r)(1, r^{-1}) = (2, 1)$, hence $x^2(2, 1)^{-1} = (0, r^2)$ and $n := x^4(0, r^2) = (4, r^2)$, a central element of order 2. In $K/\langle n \rangle$ the image of x has order 8 (as $x^4 = (4, 1) \notin \langle n \rangle$), the image of y has order 2, and $yxy = (1, r^{-1}) = (5, r)(4, r^2) = x^5 n \equiv x^5$. So $K/\langle n \rangle$, which has order 16 because $K = \{(i, r^j s^k) : i \equiv j \pmod 2\}$ has order 32, is generated by two elements satisfying the defining relations of M_{16}, whence $K/\langle n \rangle \cong M_{16}$ and $M_{16} \in HS(\mathbb{Z}_8 \times D_8) \subseteq V \vee V'$. The group M_{16} is subdirectly irreducible, its unique minimal normal subgroup being $\langle a^4 \rangle$, and it is in neither V (non-abelian) nor V' (exponent 8, while V' has exponent 4). The verbal subgroups are $\theta_V(M_{16}) = M'_{16} M^8_{16} = \langle a^4 \rangle$ and $\theta_{V'}(M_{16}) \supseteq M^4_{16} = \langle a^4 \rangle$, with equality because $M_{16}/\langle a^4 \rangle \cong \mathbb{Z}_4 \times \mathbb{Z}_2 \in V'$. Thus condition (i) fails for $A = M_{16}$, and by Theorem 1.2 the quotient map $M_{16} \to \mathbb{Z}_4 \times \mathbb{Z}_2$ lies in $W_V \cap W_{V'}$ but not in $W_{V \vee V'}$. Groups are congruence permutable, so it is distributivity that is missing.

Example 4.4 (distributive, not permutable)**.** Let $\mathcal{V}$ be the variety of bounded lattices $(L, \wedge, \vee, 0, 1)$ with an additional constant c, which is congruence distributive, and let V and V' be the subvarieties defined by $c = 0$ and by $c = 1$. Let $B = \mathbf{2} \times \mathbf{2}$ with $c = (0, 1)$; it is the product of a V-algebra and a V'-algebra, so $B \in V \vee V'$, and $\theta_V(B) = \mathrm{Cg}((0, 1), (0, 0))$ is the kernel of the first projection while $\theta_{V'}(B) = \mathrm{Cg}((0, 1), (1, 1))$ is the kernel of the second. Let $C = \{(0, 0), (0, 1), (1, 1)\}$ be the three-element chain, a subalgebra of B containing c. Then $\theta_V(C)$ identifies $(0, 0)$ with $(0, 1)$ and $\theta_{V'}(C)$ identifies $(0, 1)$ with $(1, 1)$, so the inclusion $g\colon C \hookrightarrow B$ induces bijections $C/\theta_V(C) \to B/\theta_V(B)$ and $C/\theta_{V'}(C) \to B/\theta_{V'}(B)$, both two-element lattices, while g is not surjective: C is a V-dense and V'-dense proper subalgebra, (ii) fails, and $g \in W_V \cap W_{V'} \setminus W_{V \vee V'}$. Condition (i) holds for all algebras of $V \vee V'$ by distributivity. The verbal congruences of C do not permute: $(1, 1)$ and $(0, 0)$ are related by $\theta_V(C) \vee \theta_{V'}(C)$ but not by $\theta_V(C) \circ \theta_{V'}(C)$, and C is a proper subalgebra of its pullback, which is B.

Example 4.5 (non-permuting verbal congruences, equality holds)**.** Let $\mathcal{V}$ be the variety of mono-unary algebras (A, f) with $f^4 = f^2$, and $V\colon f^2x = x$, $V'\colon f^2x = fx$. The identities $f^px = f^qx$ true in both are those with $p \equiv q \pmod 2$ and $p, q \geq 1$ (or $p = q$), so $V \vee V'$ is defined by $f^3x = fx$. An algebra B of $V \vee V'$ consists of its *core* $f(B)$, on which f is an involution, and *hairs* $x \notin f(B)$ with $f(x) \in f(B)$. The congruence $\theta_V(B)$, generated by the pairs (f^2x, x), identifies each hair x with the core element f^2x and nothing else; $\theta_{V'}(B)$, generated by the pairs (f^2x, fx), collapses each 2-cycle of the core to a point and nothing else. Hence $\theta_V(B) \cap \theta_{V'}(B) = \Delta$, which is (i). For (ii), let $C \subseteq B$ be V-dense and V'-dense. Density for V means that $C/\theta_V(C)$, which is the core of C, maps onto the core of B, so $C \supseteq f(B)$; density for V' means that the hairs of C are all the hairs of B, since hairs are not identified by $\theta_{V'}$. Thus $C = B$, and $W_{V \vee V'} = W_V \cap W_{V'}$ by Theorem 1.2. Yet the verbal congruences do not permute on the three-element algebra $A = \{0, 1, 2\}$ with $f(0) = 1$, $f(1) = 2$, $f(2) = 1$: $\theta_V(A)$ identifies 0 with 2, $\theta_{V'}(A)$ identifies 1 with 2, and $(1, 0) \in \theta_{V'} \circ \theta_V$ but $(1, 0) \notin \theta_V \circ \theta_{V'}$; A is a proper subalgebra of its pullback $A/\theta_V \times A/\theta_{V'}$, which has four elements and in which A is not V'-dense.

Remark 4.6. Condition (i) for all $A \in V \vee V'$ says that $V \vee V'$ consists of subdirect products of algebras of V and of V', which Kowalski and Paoli [8] study for disjoint subvarieties; in the congruence distributive case it holds for all pairs by Jónsson's lemma. Condition (ii) is a statement about dense subalgebras which permutability implies but which, by Example 4.5, is weaker. Theorem 1.4 gives a necessary condition for the lattice embedding in terms of $\mathrm{Sub}(\mathcal{V})$ alone, distributivity; it is not sufficient, since the variety of Example 4.4 is congruence distributive, so that its subvariety lattice is distributive by Jónsson's lemma, while (ii) fails there. No characterization of the pairs (V, V') with $W_{V \vee V'} = W_V \cap W_{V'}$ in terms of $\mathrm{Sub}(\mathcal{V})$ alone is possible: Proposition 4.7 below gives two congruence distributive varieties

with isomorphic subvariety lattices, in which (i) holds for all pairs, whose corresponding pairs behave differently, so that it is condition (ii), a statement about subalgebras, that the lattice does not see. In the language of Theorem 1.4, Example 4.3 says that $\mathrm{var}(M_{16})$ is not join-prime in the lattice of varieties of groups.

Proposition 4.7 (the condition is not lattice-theoretic)**.** *Whether* $W_{V\vee V'} = W_V \cap W_{V'}$ *holds is not determined by the position of* V *and* V' *in* $\mathrm{Sub}(\mathcal{V})$*, nor by the isomorphism type of* $\mathrm{Sub}(\mathcal{V})$*: there are congruence distributive varieties* $\mathcal{V}_1$*,* $\mathcal{V}_2$ *with* $\mathrm{Sub}(\mathcal{V}_1) \cong \mathrm{Sub}(\mathcal{V}_2) \cong \mathbf{2}\times\mathbf{2}$ *such that the equality holds for the two atoms of* $\mathrm{Sub}(\mathcal{V}_1)$ *and fails for the two atoms of* $\mathrm{Sub}(\mathcal{V}_2)$*.*

Proof. Let A_0 and A_1 be the two-element modal algebras with $\Box 0 = 0$ and with $\Box 0 = 1$, so that $\mathrm{var}(A_0)$ is defined by $\Box x = x$ and $\mathrm{var}(A_1)$ by $\Box x = 1$; each is term-equivalent to Boolean algebras, so its only subdirectly irreducible member is A_i, and neither contains the other. Put $\mathcal{V}_1 = \mathrm{var}(A_0 \times A_1)$. It has a Boolean reduct, hence is arithmetical, and by Jónsson's lemma its subdirectly irreducible members are A_0 and A_1; so $\mathrm{Sub}(\mathcal{V}_1) = \{\mathrm{triv}, \mathrm{var}(A_0), \mathrm{var}(A_1), \mathcal{V}_1\} \cong \mathbf{2} \times \mathbf{2}$, and the equality holds for the atoms by Corollary 1.3. (In the language of Section 5, $\mathrm{var}(A_0)$ and $\mathrm{var}(A_1)$ are the algebras of the logics **Triv** and **Ver**, and $\mathcal{V}_1$ those of their intersection.)

For $\mathcal{V}_2$ take the variety generated by the algebra $B = \mathbf{2}\times\mathbf{2}$ with $c = (0, 1)$ of Example 4.4. It is congruence distributive, so its subdirectly irreducible members lie in $HS(B)$. The subalgebras of B containing 0, 1 and c are the chain C and B itself; neither is subdirectly irreducible, B being a product and C having two congruences with trivial intersection, and their nontrivial quotients are the two-element algebras $\mathbf{2}_{c=0}$ and $\mathbf{2}_{c=1}$. Hence $\mathrm{Sub}(\mathcal{V}_2) = \{\mathrm{triv}, \mathrm{var}(\mathbf{2}_{c=0}), \mathrm{var}(\mathbf{2}_{c=1}), \mathcal{V}_2\} \cong \mathbf{2} \times \mathbf{2}$, the atoms being the distributive bounded lattices with $c = 0$ and with $c = 1$. The reflections of C and of B into the atoms are the two-element quotients computed in Example 4.4, so C is a proper subalgebra of B dense for both atoms, (ii) fails, and the equality fails for the atoms of $\mathrm{Sub}(\mathcal{V}_2)$. □

5. Lattices of logics and of λ-theories

For an algebraizable logic $\mathcal{L}$ with equivalent algebraic semantics the variety $\mathcal{V}_\mathcal{L}$, the lattice of axiomatic extensions of $\mathcal{L}$ is dually isomorphic to $\mathrm{Sub}(\mathcal{V}_\mathcal{L})$, an extension L corresponding to the subvariety $\mathcal{V}_L$ of algebras validating its theorems [1]. Write $W_L := W_{\mathcal{V}_L}$, the class of homomorphisms of $\mathcal{V}_\mathcal{L}$-algebras inverted by the reflection into $\mathcal{V}_L$; $L \leq L'$ gives $W_L \subseteq W_{L'}$.

Corollary 5.1. *Let* $\mathcal{L}$ *be an algebraizable logic whose equivalent algebraic semantics* $\mathcal{V}_\mathcal{L}$ *is congruence distributive and congruence permutable. Then* $L \mapsto W_L$ *is a lattice embedding of the lattice of axiomatic extensions of* $\mathcal{L}$ *into* $\mathrm{Same}(\mathcal{V}_\mathcal{L})$*:* $W_{L\wedge L'} = W_L \cap W_{L'}$ *and* $W_{L\vee L'} = W_L \vee W_{L'}$*. This applies to*

(1) *the lattice* $\mathrm{NExt}(\mathbf{K})$ *of normal modal logics, with* $\mathcal{V}_{\mathcal{L}}$ *the modal algebras* [4]*;*
(2) *the lattice of intermediate logics, with Heyting algebras;*
(3) *the lattice of substructural logics over the full Lambek calculus* **FL***, with* **FL***-algebras, and its intervals of fuzzy logics over* **MTL***,* **BL***, Łukasiewicz and Gödel logic, with* **MTL***-,* **BL***-,* **MV***- and Gödel algebras.*

Proof. Theorem 1.1 and Corollary 1.3 with $\mathcal{V} = \mathcal{V}_{\mathcal{L}}$, using the dual isomorphism $L \mapsto \mathcal{V}_L$; the meet of logics is the join of varieties and conversely. Each of the varieties listed has a lattice reduct, hence is congruence distributive, and a Mal'cev term: for modal algebras, which have a Boolean reduct, $x \oplus y \oplus z$ with $\oplus$ symmetric difference; for Heyting algebras and their expansions $p(x, y, z) = ((x \to y) \to z) \wedge ((z \to y) \to x) \wedge (x \vee z)$, which satisfies $p(x, x, z) = z$ because $z \leq (z \to x) \to x$ and $p(x, z, z) = x$ because $x \leq (x \to z) \to z$; for **FL**-algebras $p(x, y, z) = x/(z\backslash y \wedge 1) \wedge z/(x\backslash y \wedge 1)$ [7, 2, 5], which also serves all their subvarieties. □

Remark 5.2. In the framework of [14], where $(\mathrm{Same}(C), \subseteq)$ is read as a Kripke frame whose worlds are the levels of sameness, Corollary 5.1 says that the lattice of normal modal logics sits inside the frame of modal algebras as a sublattice: the logics themselves are levels of sameness for the algebras of the base logic.

Corollary 5.3. *Let* $\lambda\mathcal{T}$ *be the lattice of* λ*-theories and* **LAA** *the variety of lambda abstraction algebras. The assignment* $T \mapsto W_T := W_{\mathcal{V}_T}$*, where* $\mathcal{V}_T$ *is the subvariety generated by the term algebra* Λ/T*, is an injective map* $\lambda\mathcal{T} \to \mathrm{Same}(\mathbf{LAA})$ *preserving order and binary joins:* $W_{T \vee T'} = W_T \vee W_{T'}$*; and always* $W_{T \wedge T'} \subseteq W_T \cap W_{T'}$*.*

Proof. By [10] the lattice $\lambda\mathcal{T}$ is isomorphic to the lattice of equational theories of **LAA**, i.e. dually isomorphic to $\mathrm{Sub}(\mathbf{LAA})$, by $T \mapsto \mathcal{V}_T$; joins of theories correspond to meets of varieties, and Theorem 1.1 applies. □

Corollary 5.4. *The map* $T \mapsto W_T$ *is not a lattice embedding: there are* λ*-theories* T, T' *with* $W_{T \wedge T'} \subsetneq W_T \cap W_{T'}$*. For every pentagon* $T_0 \subsetneq T_1$*,* T' *in* $\lambda\mathcal{T}$ *with* $T_0 \vee T' = T_1 \vee T'$ *and* $T_0 \wedge T' = T_1 \wedge T'$*, the pair* (T_1, T') *is such a pair.*

Proof. The lattice $\lambda\mathcal{T}$ is not modular [11], so by Dedekind's theorem it contains a pentagon $T_0 \subsetneq T_1, T'$ as stated (Example 5.7 gives one explicitly). Under the dual isomorphism $\lambda\mathcal{T}^{\mathrm{op}} \cong \mathrm{Sub}(\mathbf{LAA})$ it becomes the pentagon $P = \mathcal{V}_{T_1} \subsetneq Q = \mathcal{V}_{T_0}$, $R = \mathcal{V}_{T'}$ with $P \vee R = Q \vee R$ and $P \wedge R = Q \wedge R$, and Theorem 1.4 gives $W_{P \vee R} \neq W_P \cap W_R$, i.e. $W_{T_1 \wedge T'} \neq W_{T_1} \cap W_{T'}$; the inclusion $\subseteq$ always holds. □

Lemma 5.5. *Let* S, T *be* λ*-theories and* Λ/S *the term algebra of* S*. The congruence lattice of* Λ/S *is isomorphic to the interval* $\{R \in \lambda\mathcal{T} : R \supseteq S\}$

of $\lambda\mathcal{T}$, a quotient Λ/R lies in $\mathcal{V}_T$ exactly when $R \supseteq T$, and the $\mathcal{V}_T$-reflection of Λ/S is the quotient map $\Lambda/S \to \Lambda/(S \vee T)$.

Proof. The first statement is the congruence correspondence for the term algebra: the congruences of Λ/S are the congruences of Λ containing S, i.e. the λ-theories containing S [10, 9]. The second follows from Salibra's dual isomorphism $T \mapsto \mathcal{V}_T = \mathrm{var}(\Lambda/T)$ between $\lambda\mathcal{T}$ and $\mathrm{Sub}(\mathbf{LAA})$ [10], used in Corollary 5.3: if $\Lambda/R \in \mathcal{V}_T$ then $\mathcal{V}_R \subseteq \mathcal{V}_T$, so $R \supseteq T$; conversely Λ/R is a quotient of Λ/T when $R \supseteq T$. Hence the least congruence of Λ/S with quotient in $\mathcal{V}_T$ is the least λ-theory containing S and T. □

Proposition 5.6. *Let $T_0 \subsetneq T_1$, T' be a pentagon in $\lambda\mathcal{T}$ as in Corollary 5.4. The quotient map $g\colon \Lambda/T_0 \to \Lambda/T_1$ lies in $W_{T_1} \cap W_{T'}$ but not in $W_{T_1 \wedge T'}$; equivalently, the term algebra Λ/T_0 of $\mathcal{V}_{T_1 \wedge T'} = \mathcal{V}_{T_1} \vee \mathcal{V}_{T'}$ is not a subdirect product of its $\mathcal{V}_{T_1}$- and $\mathcal{V}_{T'}$-reflections, so that condition (i) of Theorem 1.2 fails for the pair $(\mathcal{V}_{T_1}, \mathcal{V}_{T'})$.*

Proof. By Lemma 5.5 the $\mathcal{V}_{T_1}$-reflection of g is $\Lambda/(T_0 \vee T_1) \to \Lambda/(T_1 \vee T_1)$, the identity of Λ/T_1; its $\mathcal{V}_{T'}$-reflection is $\Lambda/(T_0 \vee T') \to \Lambda/(T_1 \vee T')$, an isomorphism because $T_0 \vee T' = T_1 \vee T'$; and its $\mathcal{V}_{T_1 \wedge T'}$-reflection is g itself, because $T_1 \wedge T' = T_0 \wedge T' \subseteq T_0 \subseteq T_1$, and g is not an isomorphism because $T_0 \neq T_1$. For the second statement, the kernels of the two reflections of Λ/T_0 are the congruences T_1 and $T_0 \vee T' = T_1 \vee T'$, whose intersection is $T_1 \neq T_0$. □

Example 5.7. Severi and de Vries [12, Thm. 44] exhibit a pentagon in $\lambda\mathcal{T}$. Let T_R be the λ-theory equating two terms when they have the same Berarducci tree, let $\Theta = (\lambda xy.y(xxy))(\lambda xy.y(xxy))$ be Turing's fixed-point combinator and $F = \Theta(\lambda x.xK)$, whose Berarducci tree is the infinite left spine $((\cdots K)K)K$, and let $\Omega = (\lambda x.xx)(\lambda x.xx)$. Then

$$T_0 = T_R{+}\{\Omega = FI\}, \qquad T_1 = T_R{+}\{\Omega = FI,\ \Omega = \Omega I\}, \qquad T' = T_R{+}\{\Omega = F,\ \Omega = \Omega K\}$$

satisfy $T_0 \subsetneq T_1$, $T_0 \vee T' = T_1 \vee T' = T_R + \{\Omega = F,\ \Omega = \Omega I,\ \Omega = \Omega K\}$ and $T_0 \wedge T' = T_1 \wedge T' = T_R$, all five theories being consistent; in the notation of [12, Thm. 44], T_0, T_1 and T' are their T_2, T_4 and T_3, and Θ is their Fix. By Proposition 5.6 the quotient map $\Lambda/T_0 \to \Lambda/T_1$, which identifies Ω with ΩI, is inverted by the reflections into $\mathcal{V}_{T_1}$ and $\mathcal{V}_{T'}$ but not by the reflection into $\mathcal{V}_{T_R}$: $W_{T_R} \subsetneq W_{T_1} \cap W_{T'}$. (The non-modularity of $\lambda\mathcal{T}$ was first proved in [11] with a different counterexample.)

Proposition 5.8. *Let $T_0 \subsetneq T_1$, T' be a pentagon in $\lambda\mathcal{T}$ as in Corollary 5.4, and let S be a completely meet-irreducible λ-theory with $T_0 \subseteq S$ and $T_1 \not\subseteq S$. Then the term algebra Λ/S is a subdirectly irreducible lambda abstraction algebra lying in $\mathcal{V}_{T_1 \wedge T'}$ but in neither $\mathcal{V}_{T_1}$ nor $\mathcal{V}_{T'}$, and the quotient map*

$$\Lambda/S \longrightarrow \Lambda/\big((S \vee T_1) \cap (S \vee T')\big)$$

lies in $W_{T_1} \cap W_{T'}$ but not in $W_{T_1 \wedge T'}$. Such theories S exist: for an equation $e \in T_1 \setminus T_0$, the λ-theories containing T_0 but not e form a nonempty family

closed under unions of chains, and any maximal member is completely meet-irreducible.

Proof. By Lemma 5.5, Λ/S is subdirectly irreducible when S is completely meet-irreducible (its congruence lattice then has a least nonzero element), $\Lambda/S \in \mathcal{V}_T$ iff $S \supseteq T$, and the $\mathcal{V}_T$-reflection of Λ/S is $\Lambda/(S \vee T)$. Now $S \supseteq T_0 \supseteq T_0 \wedge T' = T_1 \wedge T'$, $S \not\supseteq T_1$ by choice, and $S \not\supseteq T'$ since otherwise $S \supseteq T_0 \vee T' = T_1 \vee T' \supseteq T_1$. The theories $S \vee T_1$ and $S \vee T'$ therefore both lie strictly above S, so their intersection R lies strictly above S as S is completely meet-irreducible, and the quotient map $g\colon \Lambda/S \to \Lambda/R$ is not an isomorphism. Its $\mathcal{V}_{T_1}$-reflection is $\Lambda/(S \vee T_1) \to \Lambda/(R \vee T_1)$, an isomorphism because $R \vee T_1 = S \vee T_1$; likewise for T'; so $g \in W_{T_1} \cap W_{T'}$. Its $\mathcal{V}_{T_1 \wedge T'}$-reflection is g itself, as S and R contain $T_1 \wedge T'$; so $g \notin W_{T_1 \wedge T'}$. For the existence of S: the family is nonempty as T_0 belongs to it, the union of a chain of λ-theories is a λ-theory and omits e if each member does, and a maximal member S is completely meet-irreducible because every λ-theory strictly containing S contains e, so the intersection of all of them is not S. □

Remark 5.9. Proposition 5.6 is the direction "⇐" of Theorem 1.4 made concrete: the term algebra Λ/T_0 is the algebra of $V \vee V'$ that is not a subdirect product of its two reflections. Proposition 5.8 is the direction "⇒": the subdirectly irreducible algebra outside $V \cup V'$. For the pentagon of Example 5.7 the first is explicit, while the second still requires a completely meet-irreducible theory containing $T_R + \{\Omega = FI\}$ but not $\Omega = \Omega I$, for which we know no explicit description.

$$\mathrm{NExt}(\mathbf{K}) \xhookrightarrow{\text{lattice}} \mathrm{Same}(\mathbf{MA}) \qquad \mathrm{Ext}(\mathbf{FL}) \xhookrightarrow{\text{lattice}} \mathrm{Same}(\mathbf{FL})$$

$$\underset{\vee\text{-semilattice, not lattice}}{\lambda\mathcal{T}} \hookrightarrow \mathrm{Same}(\mathbf{LAA}) \qquad \underset{\text{lattice},S\text{ finite}}{\mathrm{Top}(S)} \hookrightarrow \mathrm{Same}(\mathrm{Presh}(S))$$

FIGURE 3. Lattices of theories inside sameness lattices: logics (Corollary 5.1), λ-theories (Corollaries 5.3 and 5.4), Grothendieck topologies [17].

FUNDING

No funding was received for this work.

DECLARATIONS

CRediT authorship contribution statement. Ryota Shibusawa: Conceptualization, Methodology, Formal analysis, Investigation, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.
Competing interests. The author declares that he has no competing interests.

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